From Structure to Substance: A Network and Thematic Analysis of the Hip-Hop Lyrical Universe

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Abstract

This project investigates the relationship between social structure and thematic content within the hip-hop music industry. By synthesizing network analysis and natural language processing, this study aims to determine whether artist collaborations align with distinct lyrical themes. The methodology involved a 1.5-degree data crawl using the Genius API, starting from 14 seed artists representing diverse sub-genres, resulting in a network of 180 artists and a corpus of 700 songs. Network analysis, employing centrality measures and Louvain community detection, identified seven distinct social communities. Concurrently, Latent Dirichlet Allocation (LDA) topic modeling revealed five primary lyrical themes, ranging from "Luxury/Vibe" to "Conscious/Storytelling." A comparative synthesis using statistical analysis (Chi-Squared test, $p < 6.36 \times 10^{-71}$) demonstrated a statistically significant correlation between an artist's social community and their dominant lyrical topics. The findings confirm that the hip-hop industry is stratified into distinct "nations" defined by both social connections and shared lyrical DNA, challenging the notion of a monolithic genre.

1 Introduction

The hip-hop music industry is often characterized by its polarization. Fans and critics alike divide the genre into distinct sub-cultures: the "lyrical" purists of the East Coast, the "trap" innovators of Atlanta, and the "melodic" superstars who dominate the pop charts. However, these distinct communities are typically defined by subjective labels or geographic origins rather than empirical data. This raises a fundamental sociological question: Is the hip-hop industry merely a collection of social cliques, or do these social circles actually enforce a shared "lyrical DNA"?

Previous research in Music Information Retrieval (MIR) has focused largely on audio signal processing or isolated lyrical sentiment analysis. Rarely has the relationship between the *social structure* of artists (who collaborates with whom) and the *semantic content* of their work (what they talk about) been quantified simultaneously.

This project addresses this gap by synthesizing Graph Theory and Natural Language Processing (NLP). We hypothesize that the hip-hop industry exhibits strong *homophily*—the tendency for individuals to associate with similar others. Specifically, we posit that artists do not collaborate randomly; they form distinct "nations" where social connection is a strong predictor of lyrical content. By constructing a collaboration network of 180 artists and analyzing the lyrics of 700 songs using Latent Dirichlet

Allocation (LDA), this paper attempts to map the ”Seven Nations” of hip-hop and identify the ”Bridge Artists” who possess the unique ability to traverse these thematic divides.

2 Related Work

This study builds upon three distinct pillars of computational social science and musicology: network homophily, community detection, and topic modeling.

First, the sociological foundation of this project is the concept of homophily. McPherson et al. (2001) established that similarity breeds connection, a principle observed across varied social networks [1]. In the context of music, this suggests that artists are likely to collaborate with peers who share similar artistic values and vocabularies, creating ”echo chambers” of content.

To map these social structures, we utilize the Louvain method for community detection. Blondel et al. (2008) introduced this heuristic method to optimize modularity in large networks [2]. It is particularly effective for detecting ”cliques” in social graphs, allowing us to mathematically define the boundaries between different hip-hop sub-genres without relying on subjective genre tags.

Finally, to analyze lyrical substance, we employ Latent Dirichlet Allocation (LDA). Blei et al. (2003) proposed this generative probabilistic model to discover abstract topics within a collection of documents [3]. While originally designed for text corpora like news articles, recent work by Schedl (2019) has demonstrated the efficacy of NLP techniques in Music Information Retrieval, specifically for genre classification and recommendation systems [4]. Furthermore, studies by Dodds and Danforth (2010) have successfully applied text mining to song lyrics to track temporal trends in emotional content, validating lyrics as a viable dataset for semantic analysis [5].

This project differentiates itself by combining these approaches: we use the output of the network analysis (Louvain communities) as the ground truth labels to test the distribution of the text analysis (LDA topics), thereby linking structure to substance.

3 Data Collection

To bridge the gap between social network structure and semantic content, this study required a unified dataset containing both collaboration metadata (who worked with whom) and textual data (what they said). As no pre-existing dataset offered this duality for the modern hip-hop era, we constructed a novel dataset using the **Genius API**. Genius was selected over other platforms (e.g., Spotify or Last.fm) because of its rich, crowd-sourced metadata regarding featured artists and its high-quality transcription of lyrics, including slang and ad-libs crucial to hip-hop linguistics.

3.1 Sampling Strategy: The ”Four Poles” Approach

A major challenge in network analysis is sampling bias; starting the crawl from a single artist (e.g., Drake) would result in a network heavily skewed toward pop-rap, failing to capture the genre’s diversity. To ensure a representative graph, we employed a **Multi-Seed Sampling Strategy**.

We selected 14 ”Seed Artists” acting as pillars representing four distinct sub-genres or ”poles” of the industry:

- **The Lyricists (Conscious):** Nas, Kendrick Lamar, J. Cole.
- **The Trap Innovators (South):** Young Thug, Future, Migos, Lil Baby.

- **The Melodic/Pop Bridges:** Drake, The Weeknd, Post Malone.
- **The Producers/Curators:** Kanye West, Travis Scott, Tyler, the Creator, Eminem.

This diverse initialization ensured that the resulting network would contain distinct communities rather than a single homogenous cluster.

3.2 The 1.5-Degree Network Crawl

We implemented a custom Python script to perform a "1.5-degree" breadth-first search (BFS) starting from the 14 seed artists. The crawling process followed three steps:

1. **Acquisition:** For each seed artist, we retrieved their top 50 songs sorted by popularity.
2. **Edge Extraction (Degree 1):** We parsed the metadata of these songs to identify "Featured Artists." A directed edge was created from the Main Artist to the Featured Artist (e.g., *Drake* $\rightarrow$ *Lil Baby*).
3. **Network Closure (Degree 1.5):** We did not crawl the full discography of the featured artists (which would have resulted in exponential explosion), but we included them as nodes in the network. This created a "1.5-degree" network that captures the immediate social circles of the major pillars.

3.3 Lyrical Corpus Construction

Concurrently with the network crawl, we scraped the full lyrical text for the identified songs. To ensure data quality, the raw text was passed through a filtering pipeline:

- **Length Filter:** Songs with fewer than 50 characters (instrumentals or incomplete entries) were discarded.
- **Tag Removal:** Metadata tags common in Genius transcripts (e.g., "[Verse 1]", "[Chorus]") were stripped to prevent them from influencing the topic model.

3.4 Final Dataset Statistics

The final processed dataset consists of a fully connected graph and a corresponding text corpus:

- **Nodes (Artists):** 180 unique artists.
- **Edges (Collaborations):** 326 unique connections.
- **Documents (Songs):** 700 distinct lyrical compositions.

4 Methods

To test the hypothesis of homophily, we employed a two-pronged methodological approach: structural analysis of the collaboration graph and semantic analysis of the lyrical corpus.

4.1 Network Construction

We modeled the hip-hop industry as a directed graph $G = (V, E)$, where V represents the set of 180 artists and E represents the set of 326 collaboration edges. An edge e_{ij} exists from Artist i to Artist j if Artist j was featured on a song released by Artist i. We utilized the `NetworkX` library to construct this graph, treating connections as binary (unweighted) to focus on the existence of relationships rather than their frequency.

4.2 Network Analysis Metrics

We employed two centrality measures to identify power structures within the graph:

1. Degree Centrality: This measures the total volume of connections. In our directed graph, we focused on total degree ($k_{in} + k_{out}$) to identify the ”Curators”—artists with the highest collaborative output.

2. Betweenness Centrality: This metric identifies ”Bridge” artists who act as connectors between disparate communities. It is calculated as the sum of the fraction of all-pairs shortest paths that pass through a node v:

$$C_B(v) = \sum_{s \neq v \neq t} \frac{\sigma_{st}(v)}{\sigma_{st}} \tag{1}$$

Where σ_{st} is the total number of shortest paths from node s to node t, and $\sigma_{st}(v)$ is the number of those paths that pass through v. Artists with high betweenness are theoretically positioned to transmit influence (or lyrical styles) across different sub-genres.

3. Community Detection: To identify the social ”nations,” we applied the **Louvain Algorithm**. This heuristic method partitions the network to maximize *Modularity* (Q), a measure of the density of links inside communities compared to links between communities. We utilized a standard resolution of 1.0, which successfully decomposed the network into distinct, non-overlapping clusters.

4.3 Natural Language Processing (NLP)

To analyze the ”Substance,” we processed the raw text of the 700 songs using a rigorous NLP pipeline designed specifically for hip-hop linguistics:

1. Preprocessing Pipeline:

- **Cleaning:** We removed non-lyrical markers (e.g., “[Chorus]”) and punctuation.
- **Stopword Removal:** Beyond standard English stopwords, we curated a domain-specific stopword list (e.g., “yeah”, “uh”, “yo”, “skrrt”) to prevent ad-libs from dominating the topic model.
- **Lemmatization:** Words were reduced to their base roots (e.g., “running” $\rightarrow$ “run”) using the WordNet Lemmatizer.

2. Topic Modeling (LDA): We employed **Latent Dirichlet Allocation (LDA)**, a generative probabilistic model, to discover hidden thematic structures. LDA assumes that each document (song) is a mixture of topics, and each topic is a mixture of words. We trained the model using the `Gensim` library with the following hyperparameters:

- **Number of Topics (k):** Set to 5, based on the hypothesis that lyrical themes would align roughly with the major social poles (Trap, Lyrical, Pop, etc.).
- **Passes:** 10, to ensure the model converged on stable topic distributions.

5 Findings

Our analysis revealed that the hip-hop industry is not a monolith, but a highly stratified ecosystem where social position strongly correlates with lyrical content.

5.1 The Social Structure: A Solar System of Sound

The network visualization (Fig. 1) reveals a structure analogous to a solar system. The industry revolves around a dense "Core" of super-connectors (Red/Blue nodes), while distinct sub-genres form "planetary" clusters on the periphery. The graph is fully connected, indicating that no artist is truly isolated, yet distinct communities are clearly visible.

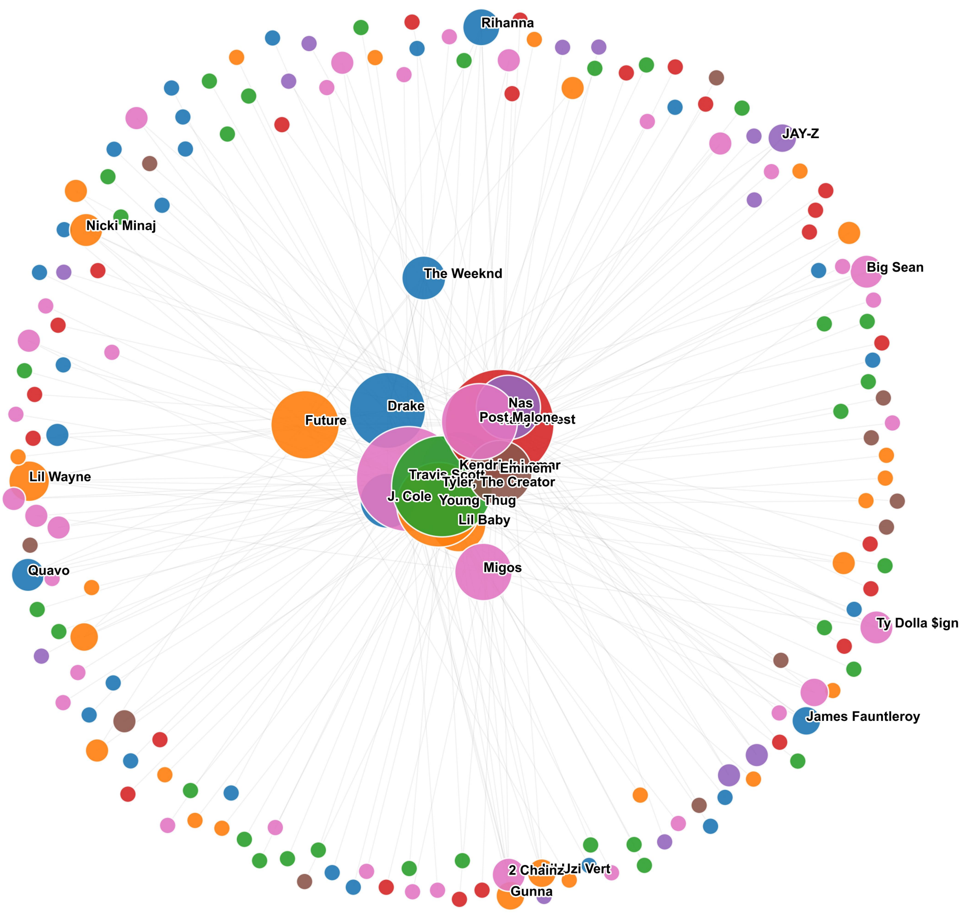

Figure 1: The Hip-Hop Collaboration Network (Colored by Louvain Community). Note the dense core versus the specialized peripheral clusters.

To better understand the structural roles, we analyzed the Centrality Metrics (Fig. 2). The scatter plot reveals a clear distinction between ”Curators” (High Degree) and ”Bridges” (High Betweenness).

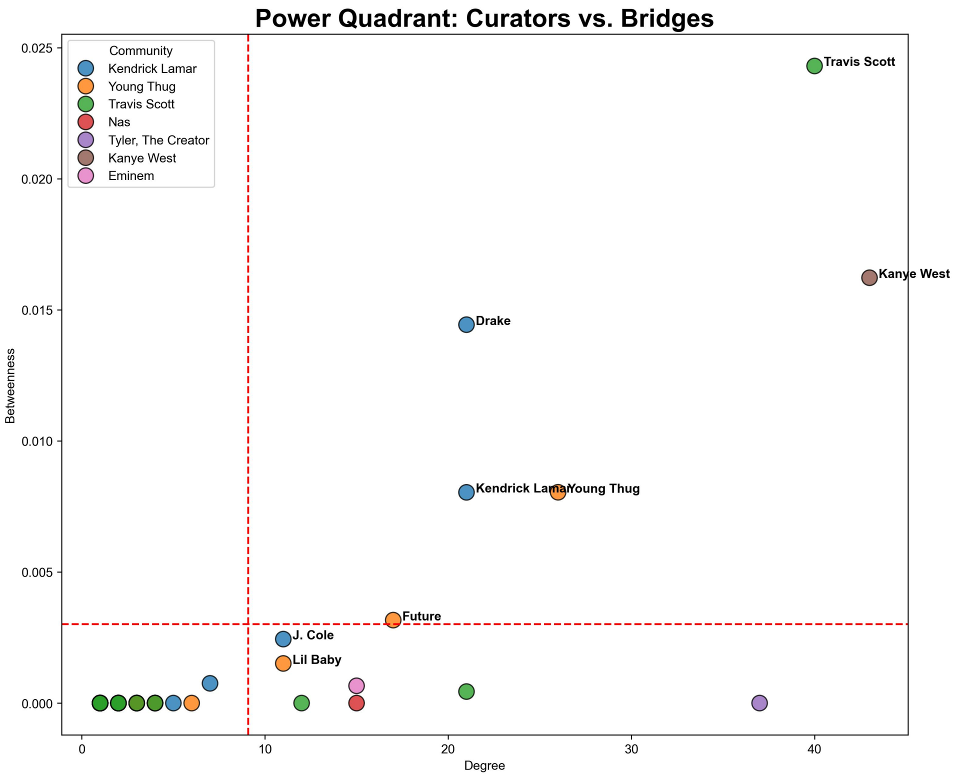

Figure 2: Power Quadrant: Degree vs. Betweenness Centrality. Travis Scott (top right) is the unique ”Bridge” connecting disparate clusters.

The bar charts (Fig. 3) confirm this hierarchy. **Kanye West** exhibits the highest Degree Centrality, confirming his role as the industry’s ”Curator”—he brings the highest volume of artists together. However, **Travis Scott** exhibits the highest Betweenness Centrality. This indicates that Travis Scott acts as the primary ”Bridge,” facilitating the flow of influence between the otherwise segregated worlds of ”Trap,” ”Pop,” and ”Lyrical” rap.

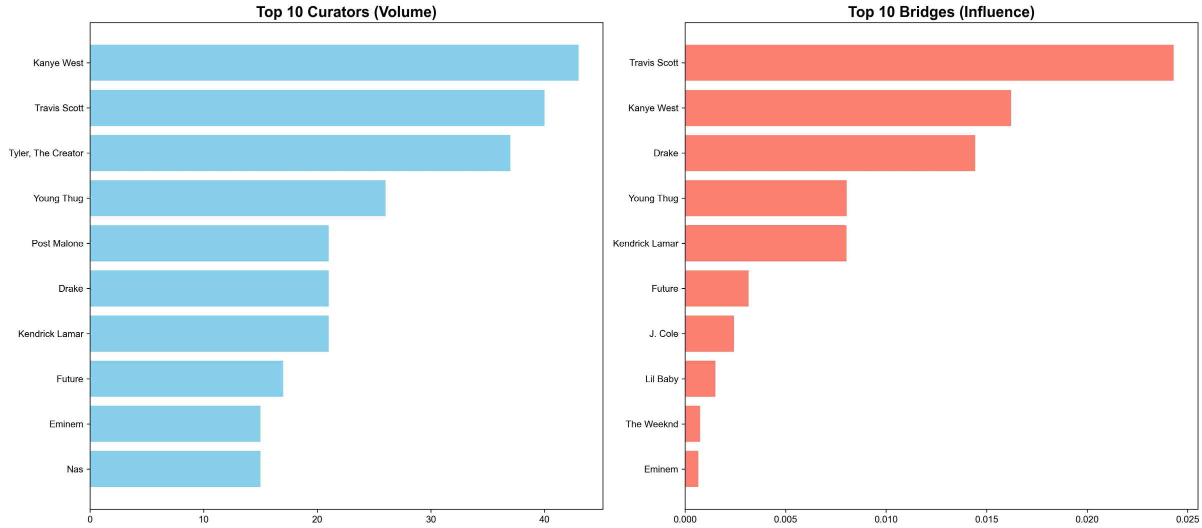

Figure 3: Top 10 Artists by Degree Centrality (Volume) and Betweenness Centrality (Bridging Influence).

5.2 Strategic Roles: Ego Network Analysis

To understand how these bridges function, we analyzed the local "Ego Networks" of key artists. We observed two distinct social strategies:

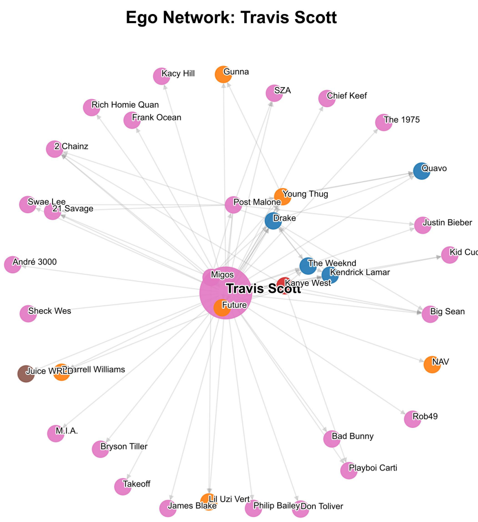

Figure 4: Travis Scott (The Bridge)

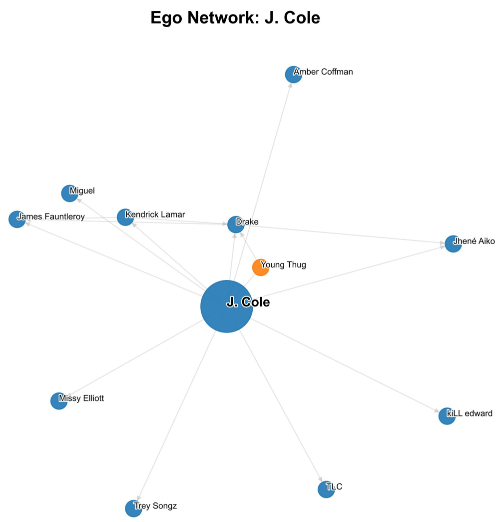

Figure 5: J. Cole (The Purist)

Figure 6: Comparison of Ego Networks. Travis Scott connects multiple colored communities, whereas J. Cole maintains a homogenous network.

As seen in Fig. 6, **Travis Scott's** network is multi-chromatic, linking pop stars (Pink nodes) with trap artists (Orange nodes). In contrast, **J. Cole's** network is sparse and homogenous, reflecting a "Purist" strategy where collaboration is reserved for peers within the same lyrical sub-genre.

5.3 The Lyrical Landscape

The LDA Topic Modeling successfully disentangled the lyrical corpus into five distinct themes, validated by the word clouds in Fig. 7:

- **Topic 0 (Luxury/Vibe):** Defined by aesthetic markers like “Versace,” “Dope,” and “Light.”
- **Topic 1 (Trap Economy):** Dominated by ad-libs and transactional language: “Money,” “Woo,” “Ayy,” “Gon.”
- **Topic 2 (Aggression):** Raw, confrontational vocabulary: “Think,” “F*ckin,” “Tell.”
- **Topic 3 (Conscious/Storytelling):** Existential and grounded themes: “Life,” “God,” “Real,” “Man.”
- **Topic 4 (Pop/Romance):** Universal themes of relationships: “Girl,” “Love,” “Baby.”

Figure 7: The Five Lyrical Themes of Hip-Hop generated by LDA Topic Modeling.

5.4 Synthesis: The Alignment of Structure and Substance

The core question of this research was whether social structure predicts lyrical content. The Heatmap (Fig. 8) provides the definitive answer. A Chi-Squared test of independence performed on the Community-Topic distribution yielded a p-value of $p < 6.36 \times 10^{-71}$, indicating a statistically significant relationship ($p < 0.05$) between an artist’s social circle and their lyrical subject matter.

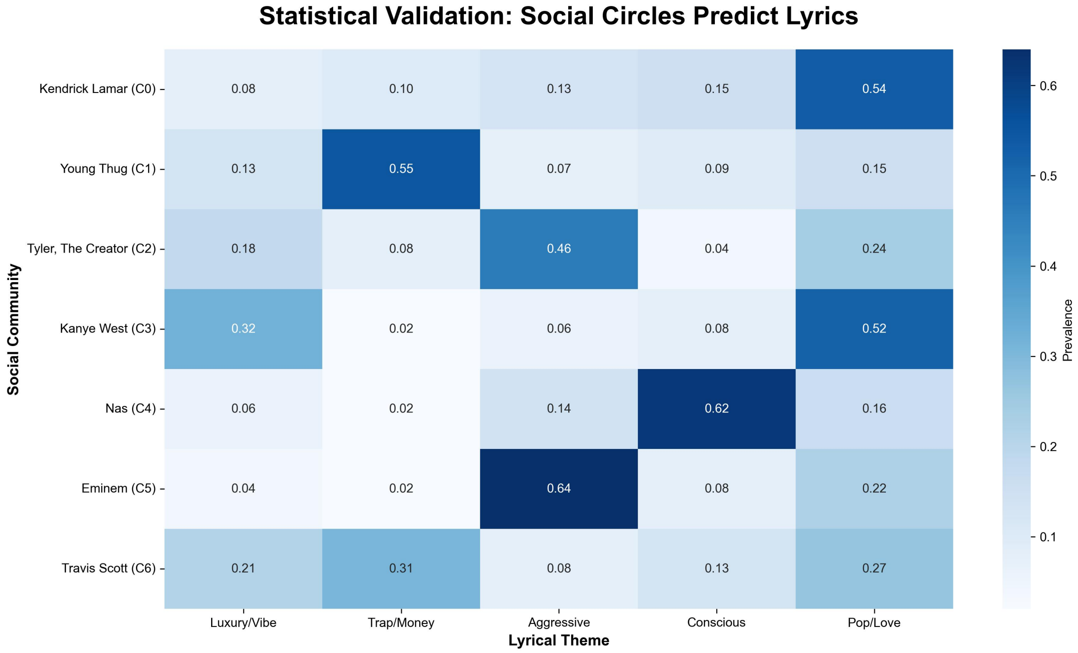

Figure 8: Heatmap showing the prevalence of lyrical topics within social communities. Darker blue indicates higher dominance.

This confirms the ”Echo Chamber” hypothesis. For example:

- **The Trap Nation (Community 1):** Led by Young Thug, this group is 55% focused on Topic 1 (Trap/Ad-libs).
- **The Conscious Nation (Community 4):** Led by Nas, this group is 62% focused on Topic 3 (Storytelling).

Finally, the Radar Charts (Fig. 9) visualize the ”fingerprints” of these nations. ”Purist” communities like Nas’s (Community 4) exhibit ”spiky” shapes, indicating hyper-specialization in one topic. In contrast, ”Bridge” communities like Travis Scott’s (Community 6) exhibit rounder shapes, indicating a fluency in multiple lyrical dialects (blending Trap and Pop).

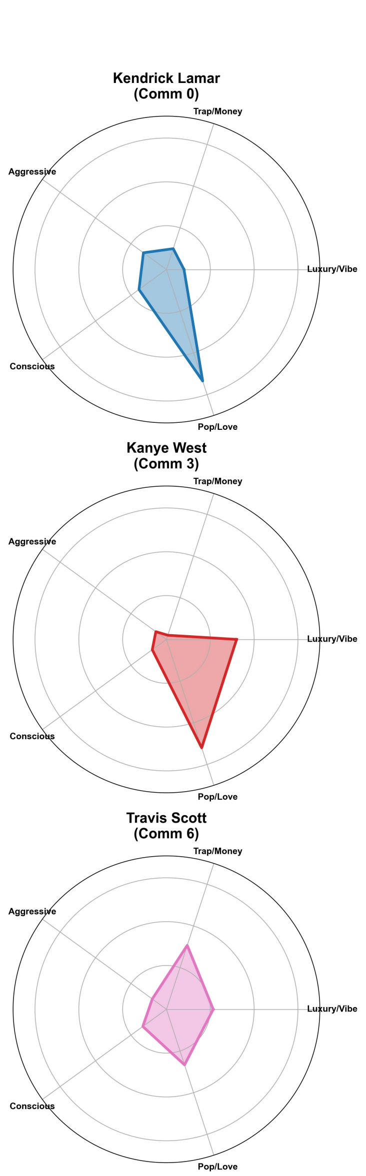

Figure 9: Lyrical "DNA" of Social Communities. Note the sharp specialization of Community 4 (Nas) versus the broader distribution of Community 6 (Travis Scott).

6 Conclusion

This study set out to answer a fundamental sociological question: Does the social structure of the hip-hop industry align with its lyrical content? By integrating network analysis with topic modeling, we have provided quantitative evidence that the answer is yes.

Our findings challenge the notion of hip-hop as a monolith. Instead, the industry functions as a "Solar System" of distinct nations. We identified that homophily is a driving force in artist collaboration; communities like the "Trap Nation" and "Conscious Nation" act as echo chambers where social isolation reinforces lyrical specialization.

Furthermore, we identified the critical role of "Bridge Artists" like Travis Scott. While "Purists" like Nas maintain the genre's thematic depth by staying within their lane, "Bridges" maintain the industry's structural integrity by connecting disparte worlds. These artists are successful precisely because they are "multilingual"—able to adapt their lyrical content to fit the social context of their collaborators.

6.1 Limitations & Future Work

While significant, this study was limited to English-language rap. Future work will incorporate multilingual datasets to analyze how these social structures function across different cultures and languages. Additionally, we aim to integrate Audio Signal Processing (e.g., MFCCs) to determine if these clusters also share a distinct ”sonic DNA” (beats and flows) alongside their lyrical themes.

References

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[2] V. D. Blondel, J. L. Guillaume, R. Lambiotte, and E. Lefebvre, “Fast unfolding of communities in large networks,” *Journal of Statistical Mechanics: Theory and Experiment*, vol. 2008, no. 10, p. P10008, 2008.

[3] D. M. Blei, A. Y. Ng, and M. I. Jordan, “Latent dirichlet allocation,” *Journal of Machine Learning Research*, vol. 3, pp. 993–1022, 2003.

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Appendix

A.1 Generative AI Usage

Generative AI (Gemini) was used to assist in debugging Python code for the Genius API connectivity, optimizing the LDA hyperparameters, and formatting the LaTeX tables.

A.2 Supplemental Materials

The full code, datasets, and high-resolution figures are available at:
[github.com/MohammedPathariya/hiphop-network-analysis]